

Allison H. Moore – Curriculum Vitae

Department of Mathematics & Applied Mathematics
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Education

Ph.D. in Mathematics, The University of Texas (2013)
B.S. in Mathematics, with High Honors, The University of Texas (2006)
B.A. in Plan II with High Honors, The University of Texas (2006)
Undergraduate Study, Independent University of Moscow, Russia (2004)

Professional Positions/Appointments

- Associate Professor, Virginia Commonwealth University
Department of Mathematics & Applied Mathematics July 2024 – current
- Assistant Professor, Virginia Commonwealth University
Department of Mathematics & Applied Mathematics August 2019 – June 2024
- Krener Assistant Professor, University of California, Davis
Department of Mathematics July 2016 -- June 2019
- Postdoctoral appointment, University of California, Davis
Department of Microbiology and Molecular Genetics July 2017-- June 2019
- RTG Lovett Instructor, Rice University,
Department of Mathematics Fall 2013 -- June 2016
- The University of Texas, Department of Mathematics:
 - Assistant Instructor Fall 2012
 - Research/Teaching Assistant Fall 2006 --- Spring 2013
 - Coordinator of the Saturday Morning Math Group Spring 2011 -- Spring 2012

Honors and Awards

- 2023, VCU, *National/International Recognition Award (NIRA) Scholar*.
- 2023, VCU, Nominated for the University Early Career Faculty Award. (Nomination only)
- 2018, University of California, Davis. *G. Thomas Sallee Mathematics Teaching Award*, \$500.
Excellence in lower division undergraduate mathematics courses.
- 2009, University of Texas. *Frank Gerth III Teaching Excellence Award*, \$1,000.
- 2013, University of Texas. Professional Development Award, \$400.
- 2009 and 2007. University of Texas. Math Department Summer Fellowships.
- 2006. University of Texas Vigre NSF Research Grant for Undergraduates, \$5,000.
- 2004. AMS/NSF Scholarship Grant, *Math in Moscow*, \$5,000.
- 2004. The University of Texas. Plan II Research and Travel Grant, \$1,500.

RESEARCH

Publications¹

Peer reviewed publications:

1. Tye Lidman and Allison H. Moore. Adjacency of Three-Manifolds and Brunnian Links. *Nagoya Mathematical Journal*. 261:e10. doi:10.1017/nmj.2025.10092, (2026).
2. Kenneth Baker, Dorothy Buck, Allison H. Moore, Danielle O'Donnol, and Scott Taylor. Primality of theta-curves with proper rational tangle unknotting number one. *Transactions of the American Mathematical Society, Series B*, 12(08):276–297, (2025).

¹ Authorship is alphabetical in Mathematics (1—23). Authorship is non-alphabetical in Education (24—25).

3. Artem Kotelskiy, Tye Lidman, Allison H. Moore, Liam Watson, and Claudius Zibrowius. Cosmetic operations and Khovanov multicurves, *Mathematische Annalen*, 389(3): 2903—2930, (2024).
4. Eugene Gorsky, Beibei Liu, Tye Lidman and Allison H. Moore. Triple Linking Numbers and Heegaard Floer Homology. *International Mathematical Research Notices*, 2023(6): 4501—4554, (2023).
5. Tye Lidman, Allison H. Moore and Claudius Zibrowius. L-space knots have no essential Conway spheres. *Geometry & Topology*, 26(5): 2065—2102, (2022).
6. Stanislav Jabuka, Beibei Liu, and Allison H. Moore. Knot graphs and Gromov hyperbolicity. *Mathematische Zeitschrift*, 301(1):811—834, (2022).
7. Christopher Flippen, Allison H. Moore, and Essak Seddiq. Quotients of the Gordian and $H(2)$ -Gordian Graphs. *Journal of Knot Theory and Its Ramifications*, 30(05):2150037, (2021).
8. Allison H. Moore and Mariel Vazquez. A note on band surgery and the signature of a knot. *Bulletin of the London Mathematical Society*, 52(6):1191—1208, (2020).
9. Eugene Gorsky, Beibei Liu and Allison H. Moore. Surgery on links of linking number zero and the Heegaard Floer d-invariant. *Quantum Topology* 11(2):323—378, (2020).
10. Allison H. Moore and Mariel Vazquez. The non-coherent band surgery model for site-specific recombination. In *Topology and Geometry of Biopolymers*, volume 746, *Contemporary Mathematics*, pages 101—125. American Mathematical Society, Providence, RI, (2020).
11. Tye Lidman, Allison H. Moore and Mariel Vazquez. Distance one lens space fillings and band surgeries on the trefoil knot. *Algebraic & Geometric Topology*; 19(5):2439--2484, (2019).
12. Kenneth Baker and Allison H. Moore. Montesinos knots, Hopf plumbings, and L-space surgeries. *Journal of the Mathematical Society of Japan*, 70(1):95—110, (2018).
13. Tye Lidman and Allison H. Moore. Cosmetic surgery in L-spaces and nugatory crossings. *Transactions of the American Mathematical Society*, 369(5):3639—3654, (2017).
14. Tye Lidman and Allison H. Moore. Pretzel knots with L-space surgeries. *Michigan Mathematical Journal*, 65(1):105--130, (2016).
15. Allison H. Moore. Symmetric unions without cosmetic crossing changes. In Gail Letzter et al., editors, *Advances in the Mathematical Sciences: Research from the 2015 Association for Women in Mathematics Symposium*, volume 6, pages 103—116. Springer International Publishing, Cham, (2016).
16. Allison H. Moore and Laura Starkston. Genus-two mutant knots with the same dimension in knot Floer and Khovanov homologies. *Algebraic & Geometric Topology*, 15(1):43—63, (2015).

Accepted Articles:

17. Matthew Elpers, Rayan Ibrahim, and Allison H. Moore. Determinants of simple theta curves and symmetric graphs. arXiv:2211.00626 [math.GT], Accepted at *Journal of Knot Theory and its Ramifications*. 12 pages.
18. Kenneth L. Baker, Allison H. Moore, Danielle O'Donnol, and Scott Taylor. Signature, slicing foams, and crossing changes of Klein graphs. arXiv:2405.15044 [math.GT], Accepted at *Algebraic & Geometric Topology*. 23 pages.

Submitted and Preprints:

19. Allison H. Moore and Nicola Tarasca. Root lattices and invariant series for plumbed 3-manifolds. arXiv:2405.14972 [math.GT], (2024). 38 pages. Submitted.
20. Allison H. Moore and Nicola Tarasca. On gluing and splitting series invariants of plumbed 3-manifolds. arXiv:2506.06515 [math.GT], (2025). 29 pages, Submitted.
21. Rayan Ibrahim and Allison H. Moore. Methods for Analyzing RNA Pseudoknots via Chord Diagrams and Intersection Graphs. arXiv:2512.19939 [q-bio.BM]. (2025). 26 pages, Submitted.

Thesis:

22. Allison H. Moore. Behavior of Knot Floer Homology Under Conway and Genus Two Mutation. *PhD Thesis*, Department of Mathematics, University of Texas at Austin, (2013).

Software and Websites:

23. Allison H. Moore, Milad Nasrollahi and Shawn Witte. *BraidGenerator* (Software), (2019). URL: <https://pypi.org/project/braidgenerator/>
24. C. Livingston and A. H. Moore, *KnotInfo: Table of Knot Invariants*, URL: knotinfo.math.indiana.edu
25. C. Livingston and A. H. Moore, *LinkInfo: Table of Link Invariants*, URL: linkinfo.sitehost.iu.edu

Scholarship of Teaching and Learning Articles:

26. [Peer reviewed conference proceeding] Bontrager, B. J., Spring, A., Satyam, V. R., Bae, C. L., Aldi, M., Moore, A. H., Tarasca, N., & Freeman, H. (2024). Instructional moves to support middle school students' engagement during a geometry camp. In K. W. Kosko, J. Caniglia, S. A. Courtney, M. Zolfaghari, & G. A. Morris (Eds.), *Proceedings of the forty-sixth annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 1767-1772). Kent State University.
27. [Book chapter] Bontrager, B., Spring, A., **Satyam, V. R.**, Bae, C. L., Aldi, M., Tarasca, N., & Moore, A. (in press). Middle school students' emotional experiences in mathematics represented through emotion graphs: A multimodal discourse analysis. In L. de Oliveira (Ed.), *Discourse Analysis in Practice* (pp. 11-30). Emerald Publishing Limited.

Grants and Funding Awarded

- NSF DMS— 2204148. *Explorations in Entanglement and Knotting in Low-Dimensional Topology*. 8/01/2022—7/31/2025, \$294,584. (Sole PI)
- NSF DMS— 2528113. *Conference: Richmond Geometry Festival: New Trends in Homological Invariants*. 8/1/2025 – 7/31/2026. \$29,276. (Co-PIs Aldi, Tarasca)
- NSF DMS— 2349810. *Conference: Richmond Geometry Festival: Geometric Topology and Moduli*. 5/1/2024 – 10/31/2025. \$26,430. (Co-PI with Aldi, Tarasca)
- NSF DMS— 2240741. *Conference: Richmond Geometry Festival: Knots, Moduli, and Strings*. 5/15/2023 – 8/15/2023. \$31,108. (Co-PI with Aldi, Tarasca)
- 4-VA Collaborative Research Grant Award (University of Virginia). \$35,000 Co-PIs with Bray, Field, Lukyanenko, Maloni, Quigley. "Collaborative workshops in topology" to be held in 2025-2026.
- VCU Breakthroughs Fund Award. *Symmetry, Surfaces, and Knots: Empowering Middle School Students*. 7/30/2022—7/30/2024, \$200,000. (Co-PI with Aldi, Tarasca)
- VCU Quest Fund Award. *Quantum Fields and Knots: An Integrative Approach*. 7/01/2022—12/31/2023, \$47,794. (Co-PI)
- Jeffress Trust Awards Program in Interdisciplinary Research—FP00014984. *Advancing theory and exploring applications of unknotting to molecular biology* August 2021 – April 2023, \$100,000. (Sole PI).
- VCU HSURP Research Project. *Knots and Links in Three-Manifolds*. Summer 2020, \$600. (Sole PI).

Other External Support and Funded Activities

- *Summer Research in Mathematics*, Simons Laufer Mathematical Sciences Institute, June 2024. Fully funded resident participant and member of research team in math biology; recipient of childcare grant.
- American Institute of Mathematics SQuaRE, *Unknotting spatial graphs*. Travel funding for five-member collaborative research project in spatial graphs. 2019-2023.
- NSF DMS—1716987. *The Dynamic Genome: Studying the Interplay between Local Strand-Passage and Reconnection*. 7/18/2017—6/30/2019, \$290,000.
Role: I co-designed the aims of the grant and co-wrote the proposal while at UC Davis with the PI (M. Vazquez). Grant provided 50% of my postdoctoral funding for two years, travel and equipment.
- NSF DMS—1148609. *RTG: Analysis, Geometry, and Topology at Rice University*, Fall 2013--Spring 2016. Role: Postdoctoral salary/travel only; no role in proposal preparation.
- NSF DMS—1148609. *RTG: Unified Training in Geometry and Topology*, Fall 2010, Summer 2011. Role: Grad student support only; no role in proposal preparation.

Presentations (From 2013 – current, for brevity)

Minicourses:

1. *Applied Topology: DNA Topology*. (Mini Course) Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, May 31, 2024.

Invited Conference Talks:

2. (i) *Series valued invariants and plumbed 3-manifolds* and (ii) *Discussion/Open Problem Session on KnotInfo and LinkInfo*. Links in Dimensions 3 and 4. ICERM, Providence, RI, May 2025
3. *Series valued invariants and plumbed 3-manifolds*. Gauge Theory and Floer Homology in Low-Dimensional Topology. Simons Center for Geometry and Physics, Stony Brook, NY Apr. 2025
4. *Root lattice puzzles and plumbed 3-manifolds*. Interactions of Geometry and Quantum Topology focused on Links in Thickened Surfaces. Banff International Research Station, Banff, Canada (Online) Apr. 2025.
5. *Tricks with Tangles*. What's your trick? A Non-Traditional Conference in Low-Dimensional Topology, Banff International Research Station, Banff, Canada, Aug. 2024.
6. *Unknotting numbers and invariants of trivalent spatial graphs*. Applied Topology: DNA topology, Material Science, Topological Data Analysis. Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, June 2024.
7. *Unknotting operations and invariants of trivalent spatial graphs*. The Crossroads of Topology, Combinatorics and Biosciences: Deciphering the Entanglement of Multi-Stranded Nucleic Acids. BIRS, Banff, Canada, March 2024.
8. *Unknotting and Invariants of Trivalent Spatial Graphs*. AMS Special Session on Mathematical Modeling of Nucleic Acid Structures. Joint Mathematics Meetings, San Francisco, CA, Jan. 2024
9. *Unknotting in Spatial Trivalent Graphs*. AMS Special Session on Recent Advances in Low-dimensional and Quantum Topology, Mobile, AL, October 2023
10. *A Tale of Two Tangles*. Tangled in Knot Theory, ICERM, Brown University, May 2023
11. *Tangle Decompositions and Immersed Curves*, Low Dimensional Topology and Circle-valued Morse Functions, IMSA, University of Miami, Feb. 2023.
12. *Invariants and unknotting in theta curves*, 8th Annual Applied Geometry and Topology Workshops in Mexico (Online), November 2022.
13. *Invariants and unknotting in theta curves*, AMS Special Session on Topology and Geometry of Multi-Stranded Nucleic Acids, Salt Lake City, UT, October 2022
14. Flash Talk: *Unknotting in spatial theta curves*, 4-variétés: à travers les dimensions, CIRM, Marseille, France, Sep. 2022
15. *Cosmetic crossings, cosmetic surgery and Conway spheres*. Canadian Mathematical Society Winter Meeting, Dec. 2021
16. *Cosmetic crossings, cosmetic surgery and Conway spheres*. Knots in Washington 49.5, Dec. 2021
17. *Tangle decompositions and immersed curves*. Boston Graduate Topology Seminar (regional conference), MIT, Boston MA, Nov. 2021
18. *Knot Graphs and Gromov Hyperbolicity*, Lightning Talk (Online), 4D Topology ARC, American Institute of Mathematics, Mar. 2021
19. *Essential Conway spheres and Floer homology via immersed curves*, Joint Meetings of the AMS/MAA, Special Session on Geometry and Topology in Dimensions 3&4 (Online), Jan. 2021
20. *Essential Conway spheres and Floer homology via immersed curves*, AMS Fall Southeastern Sectional Meeting, Special Session on Applied Knot Theory (Online), Oct. 2020
21. *Surgery, higher-order linking and Heegaard Floer homology*. Mini Symposium: Knot Theory on Okinawa, Okinawa Institute of Science and Technology, Okinawa, Japan, Feb. 2020
22. *Knot graphs and Gromov hyperbolicity*. AMS Special Session on Applications and Computations in Knot Theory, Joint Meetings of the AMS/MAA, Denver, CO, Jan. 2020
23. *Heegaard Floer invariants, surgery and Milnor Numbers*. AMS Sectional on Floer homology in Dimensions 3 and 4, University of Wisconsin-Madison, Madison, WI, Sep. 2019

24. *Site-specific recombination and the band surgery along knots*. The Topology of Nucleic Acids: Research at the Interface of Low-Dimensional Topology, Polymer Physics and Molecular Biology, Banff International Research Center, Banff, Canada, Mar. 2019
25. *Surgery on links and the d -invariant*. Winter Meeting of the Canadian Mathematical Society, Session on Topology, Vancouver, BC, Canada, Dec. 2018
26. *The band surgery model for site specific recombination along circular DNA molecules*. AMS Sectional on Mathematical Methods for the Study of the 3-Dimensional Structure of Biopolymers, SFSU, San Francisco, CA, Nov. 2018
27. *Distance one lens space fillings and band surgery*. Topology in Dimensions 3, 3.5 and 4, University of California, Berkeley, CA, May 2018
28. *Knots, bands and reconnection in circular DNA*. Eastern IL Integrated Conference in Geometry, Dynamics, and Topology, Charleston, IL, Apr. 2018
29. *Arsuaga-Vazquez Workshop Tutorials*. Summer School on Modern Knot Theory: Aspects in Algebra, Analysis, Biology, and Physics. University of Freiburg, Freiburg, Germany, Jun. 2017
30. *Cosmetic surgery in L -spaces and nugatory crossings*. Tech Topology Conference, Georgia Tech, Atlanta, GA, Dec. 2015
31. *Nugatory crossings and symmetric unions*. AMS Special Session on Knots, Links and 3-Manifolds, Rutgers, NJ, Nov. 2015
32. *Cosmetic crossing changes in thin knots*. AWM Research Symposium, University of Maryland, Baltimore, MD, Apr. 2015
33. *Cosmetic crossing changes in knots*. AMS Sectional on Knot Theory and Floer-Type Invariants, East Lansing, MI, Mar. 2015
34. *Montesinos knots, Hopf plumbings and L -space surgeries*. AMS Sectional on Interactions between Knots and Manifolds, San Francisco, CA, Oct. 2014
35. *Montesinos knots and contact structures*. AMS Sectional on Invariants in Low-Dimensional Topology, Baltimore, MD, Mar. 2014
36. *Pretzel knots admitting L -space surgeries*. AMS Sectional on Geometric Topology of Knots and 3-Manifolds, Philadelphia, PA, Oct. 2013

Invited Colloquia:

37. *Cutting Knots Down to Size: The Utility of Tangle Decompositions in Topology*. RIMS (Research Innovation in the Mathematical Sciences). University of Arizona, Tempe, AZ. Jan. 2026.
38. *Knotted graphs and graphs of knots: How combinatorics and knot theory inform each other*. In the REU Colloquium, University of Virginia, Charlottesville, VA, July. 2024
39. *Knotted graphs and graphs of knots: How combinatorics and knot theory inform each other*. University of Richmond, Richmond, VA, Oct. 2023
40. *Knots, bands and reconnection in circular DNA*. Colloquium, Virginia Commonwealth University, Richmond, VA, Dec. 2018
41. *Knots, bands and reconnection in circular DNA*. Colloquium, Western Washington University, Bellingham, WA, Nov. 2018
42. *Knots, bands and reconnection in circular DNA*. Colloquium, University of Nevada at Reno, Reno, NV, Oct. 2018
43. *Band surgery along the trefoil knot with applications to site-specific recombination on circular DNA*. Colloquium, Texas State University, San Marcos, TX, Jan. 2018
44. *Site-specific recombination in circular DNA and band surgery along the trefoil knot*. Colloquium, Towson University, Towson, MD, Dec. 2017
45. *Nugatory crossings and cosmetic surgery in L -spaces*. Colloquium, University of Alabama, Tuscaloosa, AL, Feb. 2016
46. *Montesinos knots and L -space surgeries*. Colloquium, Sam Houston State University, Huntsville, TX, Oct. 2014
47. *Mutation and total dimension in knot Heegaard Floer homology*. Colloquium, Rice University, Houston, TX, Sep. 2013

48. *Mutation and knot Heegaard Floer homology*. Colloquium, University of Nevada at Reno, Reno, NV, May 2013

Invited Seminars:

49. *Plumbed 3-manifolds and puzzles on root lattices*, Geometry Seminar, University of Virginia, Charlottesville, VA, Apr. 2025
50. *Entanglement and invariants of theta-curves*, Seminar GEOTOP-A (Online), Nov. 2023.
51. *Immersed curves in Floer homology and Conway spheres*, Topology and Geometry Seminar, MIT, Boston, MA, Nov. 2022.
52. *Cosmetic crossings*, Geometry and Topology Seminar, Le Centre de recherche en géométrie et topologie (CIRGET), Université du Québec à Montréal (Online), Apr. 2022.
53. *Conway spheres and immersed curves*. Geometry Seminar, University of Virginia, Charlottesville VA, Nov. 2021
54. *Essential Conway spheres and Floer homology via immersed curves*. Topology Seminar, Boston College, Boston MA, Nov. 2021
55. *Conway spheres and immersed curves*. Topology Seminar, Oklahoma State (Online). Nov. 2021
56. *Essential Conway spheres and Floer homology via immersed curves*, University of Texas Topology Seminar (Online) Nov. 2020
57. *Triple linking and Heegaard Floer homology*, Stanford Topology Seminar (Online) Sep. 2020
58. *Triple linking and Heegaard Floer homology*, Brandeis Topology Seminar (Online) Sep. 2020
59. *Triple linking and Heegaard Floer homology*, GA Tech Topology Seminar (Online), Aug. 2020
60. *Triple linking and Heegaard Floer homology*, The Ohio State University, Seminar on Classical Knots + Virtual Knots (Online) Aug. 2020
61. *Surgery, higher-order linking and Heegaard Floer homology*. University of British Columbia, PIMS/CRG Seminar (Online) May. 2020
62. *Heegaard Floer invariants, integral surgery and higher order linking*. Topology Seminar, Boston College, Boston, MA, Oct. 2019
63. *Heegaard Floer d -invariants and integral surgery*. Topology Seminar, Indiana University, IN, Jun. 2019
64. *Heegaard Floer d -invariants and integral surgery*. Topology Seminar, University of Oregon, Eugene, OR, May 2019
65. *Heegaard Floer d -invariants and integral surgery*. Geometry Seminar, University of Virginia, Charlottesville, VA, Apr. 2019
66. *Distance one lens space fillings and band surgery*. LA Joint Topology Seminar, UCLA, Los Angeles, CA, Apr. 2018
67. (1) *Cosmetic crossing changes* (2) *Symmetric unions without cosmetic crossing changes*, Topology Seminar(s), University of California, Berkeley, CA, Apr. 2017
68. (1) *Cosmetic crossing changes* (2) *Symmetric unions without cosmetic crossing changes*, Geometry and Topology Seminar(s), UC Davis, Davis, CA, Sep. 2016
69. *Cosmetic surgery in L -spaces and nugatory crossings*. Geometry and Topology Seminar, MSU, East Lansing, MI, Nov. 2015
70. *Cosmetic surgery in L -spaces and nugatory crossings*. Topology Seminar, UT Austin, TX, Sep. 2015
71. *Cosmetic surgery in L -spaces and nugatory crossings*. Topology Seminar, Rice, Houston, TX, Sep. 2015
72. *Cosmetic crossing changes in thin knots*. Seminar in Symplectic Geometry, Gauge Theory and Categorification Columbia University, NY, May 2015
73. *Pretzel knots admitting L -space surgeries and questions about mutation*. Geometry and Topology Seminar, Caltech, Pasadena, CA, Nov. 2013
74. *Pretzel knots admitting L -space surgeries*. Geometry and Physics Seminar, University of Miami, Miami, FL, Nov. 2013
75. *Pretzels knots admitting L -space surgeries*. Topology Seminar, Rice University, Houston, TX, Sep. 2013

Contributed and Selected Conferences

76. (Contributed) *Root puzzles and plumbed 3-manifolds*. Joint Mathematics Meetings, Seattle, WA, Jan. 2025
77. (Selected) *Essential Conway spheres and Floer homology via immersed curves*, Workshop in Geometry Topology (Online), Texas Christian University, Jun. 2021
78. (Contributed) *The cosmetic crossing conjecture and nugatory crossings*. Joint Meetings, Seattle, WA, Jan. 2016
79. (Selected) *Cosmetic surgery in L-spaces and nugatory crossings*. Workshop in Geometric Topology, TCU, Fort Worth, TX, Jun. 2015
80. (Selected) *Cosmetic crossing changes in thin knots*. Moab Topology Conference, UT, May 2015
81. (Selected) *Pretzel knots admitting L-space surgeries*. Rolfsenfest, CIRM, Luminy, France, Jul. 2013.
82. (Contributed) *The total rank conjecture in knot Heegaard Floer homology*. Joint Meetings, San Diego, CA, Jan. 2013

Other Seminars:

83. *Tait graphs & determinants of knots and spatial graphs*. Discrete MathemaTea, VCU, Oct. 2023.
84. *Root puzzles and plumbed 3-manifolds*. VCU Discrete Mathematics Seminar, VCU, Mar. 2025
85. *Introduction to Knot Floer Homology Parts 1 & 2*. VCU, Apr. 2023.
86. *Intro to Khovanov Homology (parts 1--3)* Geometry & Topology Seminar (Online), VCU, Apr. 2020
87. *Intro to Jones polynomial: skein theory, Kauffman bracket versus Jones' perspective via von Neumann algebras*. Geometry & Topology Seminar, VCU, Sep. 2019
88. *Temperley Lieb algebras*, Geometry & Topology Seminar, VCU, Sep. 2019
89. *Gordian graphs of knots*. ALPS Seminar, VCU, Sep. 2019
90. *Behavior of knot Floer homology under Conway and genus two mutation*. Topology Seminar (Thesis Defense), University of Texas, Austin, TX, Apr. 2013

TEACHING

Courses Taught as Instructor of Record

Math 697, Directed Research, VCU	F21 -- F23
Math 610, Graduate Advanced Linear Algebra, VCU	S26
Math 602, Graduate Abstract Algebra II, VCU	S22
Math 502, Graduate Abstract Algebra I, VCU	F21
Math 490, Mathematical Expositions, VCU	S23, F23
Math 411, Excursions in Geometry, VCU	F23
Math 409, General Topology, VCU	F22
Math 401, Abstract Algebra, VCU	F20, F24
Math 310, Linear Algebra, VCU	F19, F20, S21, S22, S25
Math 307, Vector Calculus, VCU	S21, S24
Math 200, Calculus Analytic Geometry 1, VCU	S20, F24
Math 170, The Language of Artificial Intelligence, VCU	F25
Mat 22A, Linear Algebra, UC Davis	S17, F17, F18
Mat 141, Euclidean and Non-Euclidean Geometry, UC Davis	W17
Mat 116A, Short Calculus, UC Davis	F16
Math 306, Elements of Abstract Algebra, Rice University	S16
Math 111, Differential Calculus, Rice University	F15
Math 445/540, Graduate Algebraic Topology, Rice University	S15
Math 499, Undergraduate Research Seminar, Rice	F14—S16
Math 591, Graduate Teaching Seminar, Rice	AY 2014—2016
Math 699, Topology Seminar, Rice	AY 2014—2016
Math 112, Integral Calculus, Rice University	S14
Math 368, Graph Theory and Combinatorics, Rice	F13, F14
M408K, Differential Calculus, University of Texas	F12

Courses taught as a Teaching Assistant at the University of Texas:

Recitations (6 hours weekly) with office hours (3 hours) and grading:

M408M, Vector Calculus	F2008, S2009
M408L, Integral Calculus	S2008, Summer 2008
M408K, Differential Calculus	F2007
M408C, Sequence, Series, and Multivariable Calculus	F2006, S2007

Office hours and grading only:

M382C, Graduate Algebraic Topology	F2009
M346, Applied Linear Algebra	F2009

New Classes Introduced:

- Math 411: Excursions in Geometry: new required course for math majors.
- Math 170: The Language of AI: general education math; CHS AI Minor & BIS Practical AI+ program.

Instructional Materials

- Math 411 (Excursions in Geometry): New Course Complete lecture notes made available to students.
- Math 502/602 (Graduate Abstract Algebra I and II): Complete lecture notes made available to students.
- Math 610 (Graduate Advanced Linear Algebra): Complete lecture notes made available to students.
- Math 310 (Linear Algebra): Videos (56 tutorials); learning modules and complete lectures (25 PDFs)
- Math 401 (Abstract Algebra): Complete lecture notes made available to students (34 PDFs)
- Math 170 (Language of AI) Lecture notes, and complete set of worksheets and activities.

Graduate Students

- Rayan Ibrahim. Ph.D. in Systems, Modeling and Analysis, VCU. Graduated August 2024; currently a Visiting Assistant Professor at Lafayette College.
- John Carney. Masters in Mathematics, VCU. Graduated August 2024
- Jonathan Gerhard. Masters in Mathematics, VCU. Degree expected May 2027.

Sponsorship of Undergraduate Research

- 2024-2026. Supervised 4 undergrads RAs on NSF-funded projects in knot theory, combinatorics and mathematical biology.
- 2022-2024. Supervised student (as both undergrad and Master's student) on a PerQ, Jeffress and NSF-funded projects in knot theory resulting in article submitted to math journal.
- 2021-2023. Supervised 5 undergrads RAs on various Jeffress Trust-funded projects in knot theory and graph theory. One project resulted in math journal publication.
- Summer 2020. Honors Summer Undergraduate Research Program (HSURP). Supervised two undergrads on a project on "Knots and Links in Three-Manifolds" resulting in math journal publication.
- Spring 2019. Advised 1 undergrad and 1 grad in computational knot theory project resulting in the Markov-chain based software *BraidGenerator*. Advised 1 undergrad on project in knot theory.
- Summer 2018. *Knot invariants and neural knots*, UC Davis. Supervised 1 undergrad in the construction of a feed-forward neural network to predict knot invariants.
- Spring 2016. *TDA & Low-Dim Topology*, Rice University. 7 students.
- Fall 2015. *Combinatorial knot theory*, Rice University. 12 students.
- Spring 2015. *Combinatorial knot theory*, Rice University. 7 students.
- Fall 2014. *Topological data analysis*, Rice University. 7 students.

Other Teaching Activities and Initiatives

- VCU Twists, Turns and Trajectories Camp, June 2-6, 2025. Math, stats and physics camp for middle school students. Teacher and co-organizer.

- VCU Geometry Camp, June 12-16, 2023. Math camp and IRB-approved education study for middle school students. Co-PI, teacher, and co-organizer.
- VCU Knots and Graphs, May 2022 – Co-organizer of an informal REU at VCU.
- Seminar Co-organizer, *Geometry Student Seminar*, VCU, Fall 2022.
- Seminar Co-organizer, *Teaching Seminar*, Rice University, Fall 2014 -- Spring 2016.
- Teacher, *RTG Writing Seminar*, Rice University, Fall 2014.
- Mentor, RTG Directed Reading Program, University of Texas, Spring 2013.
- Co-organizer, TA Training Workshop, University of Texas, Summer 2012.
- Supplemental Instruction Training, University of Texas, Fall 2007-- Spring 2009.

SERVICE

Service to the Profession

Referee Work & Expert Opinions:

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| • <i>Algebraic & Geometric Topology</i> | • <i>Mathematical Research Letters</i> |
| • <i>Advances in Mathematics</i> | • <i>Michigan Math Journal</i> |
| • <i>American Mathematical Monthly</i> | • <i>Pacific Journal of Mathematics</i> |
| • <i>BIRS Workshop Proposals</i> | • <i>Proc. of the American Math. Society</i> |
| • <i>Bulletin of the London Mathematical Society</i> | • <i>Proc. of the Edinburgh Math. Society</i> |
| • <i>Cambridge Philosophical Society</i> | • <i>Proc. of the London Math. Society</i> |
| • <i>Canadian Journal of Math</i> | • <i>Quantum Topology</i> |
| • <i>Compositio Mathematica</i> | • <i>Selecta Mathematica</i> |
| • <i>Geometriae Dedicata</i> | • <i>SIGMA (Symmetry, Integrability and Geometry: Methods and Applications)</i> |
| • <i>Geometry & Topology Monographs</i> | • <i>Springer Conference Proceedings</i> |
| • <i>Involve: A Journal of Mathematics</i> | • <i>Transactions of the American Mathematical Society</i> |
| • <i>Math. Proc. Cambridge Philosophical Society</i> | |

NSF Review Panels: Served on three panels in 2024, 2023, 2020.

Conference and Workshop Co-Organization:

- *Knots, Groups, and Manifolds*. Centre de Recherches Mathématiques, UQAM, Montreal, August 11-15, 2025.
- *4-VA Collaborative Research Grant Workshops*. To be held in Virginia in summer 2025—2026.
- *Applied Topology: DNA topology, Material Science, Topological Data Analysis Session and Mini Course*. Canadian Mathematical Society Summer Meeting, Saskatoon, Canada, May 2024.
- *Richmond Geometry Day*, VCU (*To be held*) February 7, 2026.
- *Richmond Geometry Festival V*, VCU. September 19-21, 2025.
- *Richmond Geometry Festival IV*, VCU. August 12-14, 2024.
- *Richmond Geometry Festival III*, VCU. June 2-4, 2023.
- *Richmond Geometry Festival II*, VCU. (Virtual). May 27-28, 2022.
- *Richmond Geometry Festival I*, VCU (Virtual). June 10-11, 2021.
- *AMS Special Session on Knots and Links in Low-Dimensional Topology*, AMS Sectional Meeting, University of Virginia, *cancelled due to Covid.
- *AMS Special Session Developments in Spatial Graphs I and II*, Joint Meetings of AMS/MAA Online, January 6, 2021
- Co-leader (with J. Hom) of 8-member team. *Women in Symplectic and Contact Geometry and Topology Workshop* (WiSCon), ICERM, Providence, RI, July 2019.
- Conference Tutor. *Summer School on Modern Knot Theory: Aspects in Algebra, Analysis, Biology, and Physics*. University of Freiburg, Freiburg, Germany, June 2017.
- *Topology in Dimension 3.5: A conference in memory of Tim Cochran*, Rice University, June 2016

- *AMS Special Session on Geometric Perspectives in Knot Theory*, AMS Sectional Meeting, Loyola University, Chicago, October 2015.

Other Service to the Profession:

- Develops and maintains *KnotInfo* with C. Livingston. KnotInfo: Table of Knot Invariants. Free online research database. <http://knotinfo.org/>
- Co-organizer of *Virtual Low-Dimensional Topology*. Online community that hosted a seminar, virtual office hours, and events for the topology community during Covid-19. <https://sites.google.com/bc.edu/virtualtopology/home>
- Panelist in career panel at 3rd Richmond Geometry Meeting at VCU, June 2023.
- Panelist in career panel at 4th Annual SCMB Symposium in Math-Bio, Dec 2021.
- Panelist in career panel at Eastern IL Integrated Conference in Geometry, Dynamics, and Topology, Charleston, IL, Apr. 2018
- Panelist in grad student panel at NCUWM in 2012.

Professional memberships: American Mathematical Society, Association for Women in Math (past)

Service to the University (2013 – current)

University and College Level Committees and Activities

- AAC&U Institute on AI, Pedagogy, and Curriculum team, F24—S25
- VCU CHS *Faculty Learning Community on Mentoring of Students in Research*, S24
- VCU CHS Grants Reviewer, S23.
- VCU CHS Awards, Leave & Grants Review; Ad Hoc Committee, F23—S24; F25
- VCU OVPRI Proposal Reviewer, S24

Seminar Organization

- Geometry and Topology Seminar, VCU, F19—current
- Colloquium, VCU, F21—S22, F25
- ALPS Seminar, VCU, F20—S21, F22—S23
- Topology & Working Topology Seminars, Rice University, F14—S16
- Heegaard Floer 'Computational,' Rice University, F14—S15
- Teaching Seminar, Rice University, F14 – S16.
- RTG Writing Seminar, Rice University, Fall 2014.

Math Department Committees and Activities

- *Undergraduate Research*, VCU F24—F25
- *Open House*, VCU F24—F25
- *Math Circle & Outreach*, VCU F24—F25
- *Graduate Affairs*, VCU, F23—S25
- *Undergraduate Curriculum*, VCU, F20—S22
- *Undergraduate Affairs*, VCU F25
- *Analysis Coordinator*, VCU, F22—S23, F24—S25
- *Undergraduate Research*, VCU, F22—F23
- *Website*, VCU, F21—S22
- *Covid Prep Team*, VCU, (Voluntary summer committee), Summer 2020.
- *Colloquium Committee*, Rice, F13--S14.
- *Distinguished Women in Mathematics*, University of Texas, F10--S13.

Letters of Recommendation and Career Mentorship:

- Postdoctoral Job & Fellowship Letters: 3 individuals (2023-2025)
- VCU Students: **Numerous** letters for many individuals for the following types of programs:
 - Postdoctoral and faculty positions
 - Ph.D. graduate programs
 - M.S. graduate programs
 - HSURP program
 - Scholarships and fellowships
 - Financial aid appeal
 - REUs & summer/internships
 - Transfer programs
 - Job application references
- UC Davis Students: Numerous letters for graduate programs, transfer programs, scholarship recommendations, and other. (2016—2019)
- Rice University Students: Several letters for graduate Ph.D. programs/fellowships, lecturer position (2015—2016)
- USCIS: National interest waiver petitions for 3 individuals (2019, 2021, 2023).
- Other: One letter of support for a postdoctoral scholar award (2020). One letter of support for sabbatical visitation (2021).
- Career Mentor: 3 VCU Grad Students; Numerous prior at UC Davis, Rice.
- Mentor, RTG Directed Reading Program, University of Texas, Spring 2013.

Service to the Community

- Outreach: Career Day at Bon Air Elementary, May 2023.
- VCU Math Circle Activity Leader: April 2022 and Dec. 2022.
- Math Camp Colloquium Speaker, Texas State University, July 2018.
- M-PACT middle school outreach volunteer, UC Davis, April 2018.
- STEM for Girls Workshop volunteer, UC Davis, April 2017
- Math Department Picnic Day volunteer, UC Davis, April 2017
- Faculty sponsor of the Association for Women in Math, Rice U Chapter, 2013 – 2016.
- REU Activity Leader, Sam Houston State University, July 2015
- Girls Exploring Math and Science, Houston Museum of Natural Science, February 2014.
- Panelist & Co-organizer; Sonya Kovalevsky Day, University of Texas, S13, S14.
- Coordinator/Speaker, Saturday Morning Math Group/Sunday Math Circle, 2010—2012.
- Sponsor, AMC and AIME Exams, University of Texas, F11- F12